

An outlier detection framework for Air Quality Index prediction using linear and ensemble models

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ABSTRACT

The Air Quality Index (AQI) is a key indicator for assessing air quality and its associated health impacts. Accurate AQI calculations are crucial for reliable air quality assessments, but outliers in air quality data can distort these calculations, leading to inaccurate predictions. This paper presents a comprehensive framework for air quality prediction that integrates multiple outlier detection methods with machine learning models, focusing on enhancing the accuracy and robustness of predictions. The study investigates various outlier detection techniques, including the Interquartile Range (IQR), robust Z-score, and Mahalanobis distance, and evaluates their impact when integrated into machine learning models. Unlike traditional approaches that remove outliers without considering seasonal effects, this research proposes retaining extreme data points after seasonal validation to improve model generalization and prediction accuracy for unseen data. The framework is evaluated using a dataset from Jaipur city, testing multiple machine learning models, including linear regression, ensemble methods, and K-Nearest Neighbor (KNN) regression. Results show that the integrated framework significantly improves model performance, with the Extra Trees Regressor achieving the best results (MAE = 11.9161, RMSE = 16.1660, and $R^2 = 0.8884$) after refinement, compared to baseline performance (MAE = 12.6765, RMSE = 17.8452, and $R^2 = 0.8737$). This study demonstrates the empirical effectiveness of the proposed framework and provides practical guidelines for air quality prediction in real-world applications.

1. Introduction

Air pollution has a major impact on the health of humans as well as on the environment. Pollution should be monitored continuously so that it can be controlled effectively. The data presented by the World Health Organization (WHO) is warning about the pollution's levels in the country. A recent report by Xing and Sun [1] reveals that two-thirds of the 30 most polluted cities globally are in India, with six in the top 10. Ghaziabad, a satellite of Delhi, ranks as the worst, with pollution levels nine times above what U.S. Environmental Protection Agency (EPA) considers healthy. Delhi was the world's most polluted capital city in 2023 [2]. The Air Quality Index (AQI) is a standardized indicator used globally to report daily air quality levels and their potential health effects. AQI calculations rely heavily on the concentrations of various pollutants, particularly Particulate Matter (PM) such as PM_{2.5} and PM₁₀.

Jaipur, the capital of Rajasthan in India, is famous for its rich heritage, lively culture, and unique pink-hued architecture, which has earned it the title "The Pink City". The city has a semi-arid climate, featuring hot summers, a distinct monsoon season, and mild winters. Like

many prominent cities in India, Jaipur confronts significant challenges related to air pollution. Factors such as rapid urban growth, rising vehicle traffic, and industrial operations contribute to the deterioration of air quality. During winter months, pollution levels often spike due to temperature inversions and increased use of biomass for heating.

Air pollution significantly affects both ecological balance and human health. As the concentration of gases in the atmosphere increases, it poses considerable risks to the human body. Meena et al. [3] emphasize the potential of Machine Learning models to predict and influence travel behavior based on awareness of air quality. Their study highlights the necessity for enhanced air quality management and public education to encourage healthier and more sustainable travel choices. In a related effort, Tavana et al. [4] present a comprehensive model that integrates location, inventory, and routing decisions aimed at minimizing carbon emissions in supply chains, thereby contributing to reduced air pollution. Additionally, Palit et al. [5] provide an in-depth analysis of how Electric Vehicles (EV) can help mitigate air pollution and advocate for the adoption of EVs as a sustainable transportation solution in developing economies.

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In Jaipur, PM, Nitrogen Dioxide (NO₂), and Ozone (O₃) are the most common types of air pollution. PM_{2.5} and PM₁₀ are primary indicators of air pollution due to their adverse health effects and prevalence in urban atmospheres. PM_{2.5} is particularly harmful as it penetrates deeper into the lungs and bloodstream, causing severe respiratory and cardiovascular issues. PM₁₀ also poses health risks but to a slightly lesser extent due to its larger particle size, which limits its penetration into the human body Xing and Sun [1], Colangeli et al. [2].

A thorough literature review was performed to identify studies focused on air pollution, particularly examining the machine learning methods employed and their effectiveness in producing accurate results. These systems play a crucial role in monitoring air pollutants in specific regions, enabling comprehensive air quality analysis and forecasting future trends. One notable system developed by Imam et al. [6] combines the Internet of Things (IoT) with a machine learning algorithm, specifically employing Recurrent Neural Networks (RNN) that utilize Long Short-Term Memory (LSTM) units to enhance air pollutant monitoring capabilities. Another study by Evagelopoulous et al. [7] introduces a cloud-based platform that utilizes knowledge-based discovery to deliver real-time air quality data. This information is collected from monitoring stations distributed across various geographical areas and is made accessible through mobile applications, facilitating timely and comprehensive monitoring. Dhingra et al. [8] developed a mobile application IoT-Mobair to monitor real-time air quality data. The data are collected by monitoring stations that are placed in various geo-locations. This system uses mobile clients for monitoring purposes. Sonu and Suyampulingam [9] conducted a historical air quality data analysis for Coimbatore city and implemented air quality prediction using a linear regression model. Similarly, IoT based systems proposed by Pushpam et al. [10], Niranjan and Rakesh [11], Pasupuleti et al. [12] are used to collect and monitor real time air quality data. Zhang and Woo [13] explored a new way to predict immediate air quality around people, by combining fixed and mobile sensors. The experimental results show that hybrid distributed fixed and IoT sensor systems are effective in predicting air quality around the people. Machine learning techniques are extensively utilized in numerous highly cited studies [14–18] for predicting air pollutant concentrations and the AQI. Recent research [19] shows that the machine learning model exhibited superior predictive capabilities compared to the traditional mass-balance (MB) model. The machine learning model demonstrated strong Index of Agreement (IOA > 0.69) and Pearson correlation coefficients ($r > 0.80$). Mehmood et al. [20] aims to provide insights, research priorities, and opportunities for enhancing the accuracy of air quality prediction and managing air pollution emergencies on a global scale. Wang et al. [21] employed Support Vector Regression (SVR) and Random Forest Regression (RFR) to forecast AQI and PM concentrations in Beijing. It demonstrated high predictive accuracy for AQI using these models. Hossain et al. [22] developed a system that automatically analyzes and detects pollution using AI and object detection.

Numerous studies in India [23,24] have focused on predicting air quality using machine learning techniques. Deep learning algorithms are increasingly being utilized, demonstrating promising results in forecasting the Air Quality Index (AQI). Key contributions to this research area include work by Sarkar et al. [25], Ahmed et al. [26], Rabie et al. [27], and Ravindiran et al. [28]. Notably, Wu et al. [29] utilized a LSTM neural network model to predict the PM_{2.5}/PM₁₀ ratio, which provided insights into the sources of pollutants and improved AQI predictions by effectively capturing temporal dependencies within the data.

Outlier detection plays a critical role in the domain of air quality prediction. Accurate monitoring and forecasting of air quality are essential for public health, environmental management, and policy-making. Outliers [30], which are data points significantly different from the majority of observations, can distort statistical analyses and predictive models. In air quality datasets, these anomalies may arise from sensor malfunctions, data entry errors, or rare environmental

events [31]. If not detected and handled appropriately, outliers can skew model training, leading to biased or inaccurate predictions. We can enhance the accuracy and robustness of models by scrutinizing and treating outliers within the training data. This ensures the models learn from a more representative sample of the population. Majorly we can categorize outlier detection in two categories: 1. Statistical methods, 2. Machine learning methods. Statistical methods like Z score, Interquartile Range (IQR) leverage statistical properties of the data to identify points that deviate significantly from the expected pattern. Machine learning methods utilize algorithms to learn patterns within the data and identify anomalies that deviate from those patterns.

The presence of outliers in AQI datasets can significantly skew prediction models, leading to inaccurate forecasts. A comprehensive framework is presented by Dash et al. [32] for detecting and handling outliers in datasets using the Inter-Quartile Range (IQR) method. Research by Wang et al. [21] introduced a semi-supervised broad learning system (CC-SSBS) that incorporates a correlation entropy criterion to manage outliers. This model demonstrated improved stability and accuracy in AQI predictions by reducing the impact of anomalous data. By utilizing geographic and temporal data, it better understood and mitigated the influence of outliers on AQI prediction. Chandra et al. [33] proposed handling missing and imbalanced data in AQI predictions using a Median-KNN regressor and SMOTE (Synthetic Minority Over-sampling Technique)-Tomek Links. This approach effectively addresses the issues of missing values and class imbalance, improving prediction accuracy by pre-processing the AQI dataset to handle outliers more robustly. A comprehensive review by Bellinger et al. [34] emphasized the importance of data pre-processing, including outlier detection and removal, in improving the performance of machine learning models for AQI prediction. Techniques such as support vector machines and artificial neural networks were particularly noted for their effectiveness when paired with robust pre-processing steps. Gupta et al. [18] conducted a comparison of various machine learning techniques, such as decision trees and K-nearest neighbors (KNN), to predict the Air Quality Index (AQI). Their findings indicated that models which included pre-processing steps to manage outliers – like repeated cross-validation and outlier removal – demonstrated improved accuracy and reliability in their predictions. Wang et al. [35] investigated hybrid deep learning models that integrate SVR with LSTM networks for AQI prediction. The effectiveness of these models in handling outliers was enhanced by employing advanced pre-processing techniques that clean the data prior to its input into the predictive model.

Another study by Wang et al. [21] conducted a study that utilized multimodal imputation techniques to address missing data and correct outliers. This method was shown to improve the performance of various machine learning models used for predicting the Air Quality Index (AQI), highlighting the essential role of outlier management in data pre-processing. Several studies have indicated a correlation between elevated particulate matter (PM) concentrations during the colder months, including findings by Xing and Sun [1], Saminathan and Malathy [36] and [37]. Seasonal trends were consistent, showing higher pollution levels during winter due to increased heating activities. In many regions seasons affect air quality strongly. The outlier analysis based on the domain knowledge and seasonal correlation will be key in improving AQI prediction. Recent advancements in machine learning and data pre-processing techniques have shown promise in addressing challenges in prediction in presence of outliers. By incorporating statistical methods and machine learning models, researchers can enhance the accuracy and reliability of AQI predictions, ultimately contributing to better environmental monitoring and public health outcomes.

This paper is structured into four sections. The first section reviews the current literature on air quality prediction and outlier detection. The second section details the dataset and methodology used. The third section presents the prediction accuracy of various machine learning techniques, both with and without the presence of outliers. The final discussion section draws two key conclusions: seasonal analysis should

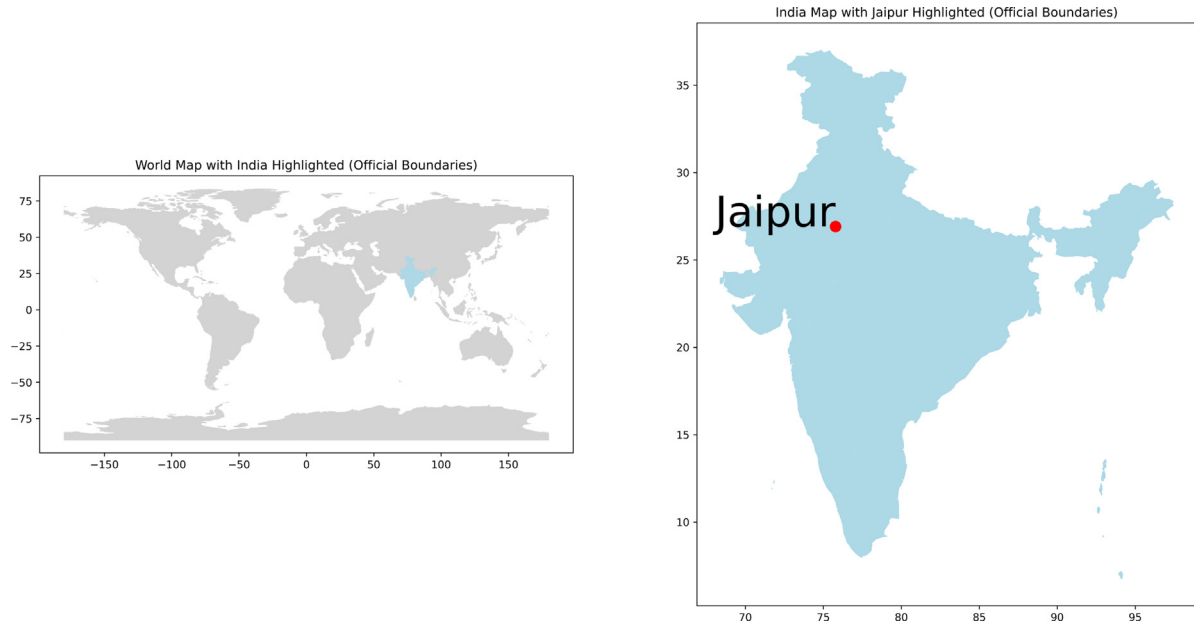


Fig. 1. Location of Jaipur.

be integrated into outlier processing, and ensemble methods demonstrate greater robustness compared to linear models in predicting the air quality of Jaipur in the presence of outliers.

The contributions of this research are summarized as follows:

- Air quality data for Jaipur city, India, was collected from the CPCB (Central Pollution Control Board).
- Several data pre-processing methods were applied to the air quality dataset to normalize the data and manage missing values.
- Outlier analysis was conducted using statistical techniques. Seasonal analysis was also performed to validate the identified outliers.
- The overall proposed architecture is depicted in Fig. 2.
- Various linear models, such as linear regression, Lasso regression, and ridge regression, along with ensemble methods like random forest, Extra Trees, and gradient boosting, were employed to train a machine learning model for predicting the AQI of Jaipur city.
- A comparative analysis of these machine learning models was conducted, demonstrating that the Extra Trees Regressor yielded superior results for AQI prediction, as evidenced by evaluation metrics such as the R^2 score and Root Mean Square Error (RMSE).
- Ensemble methods exhibit greater robustness in the presence of outliers compared to linear models.

2. Data and methods

2.1. Dataset description

This research is being conducted to observe the impact of outliers in the Indian city, Jaipur. Jaipur as shown in Fig. 1, located in the state of Rajasthan, has a population of 3.1 million people [38] and spans an area of 484.64 km². Jaipur experiences extreme temperatures, particularly during the summer and winter months, with temperatures often ranging from 4 to 45 degrees Celsius. The climate in Jaipur is characterized by a hot semi-arid environment, marked by significant temperature variations between seasons and relatively low annual precipitation. Data for this study was provided by India's Central Pollution Control Board, New Delhi [39]. The location of six air quality monitoring stations is shown in Table 1. Daily station records over a period of two years, starting on January 1, 2018, and ending on

Table 1

Jaipur air quality station locations.

Station code	Location
298	RSPCB Office, Jhalana Dongari
410	RIICO Office MIA, Jaipur
296	PHD Office, Ajmeri Gate
408	Office of the District Educational Officer, Chandpol
409	RSPCB, RO, Vidyadhar Nagar
297	VKIA, Jaipur (Road no.-6)

Table 2

Jaipur dataset description.

Dataset statistics		Variable types	
Number of variables	24	Categorical	2
Number of observations	1114	DateTime	1
Missing cells (%)	5.30%	Numeric	19
Duplicate rows	0	Unsupported	1

December 31, 2019, make up the investigated data. As a consequence, there are 720 days since every month has been linearly extrapolated to have 30 days in order to meet the criteria for converting discrete data to functional data. Both cities' PM_{2.5}, PM₁₀, NO₂, NO_x, O₃, and SO₂ levels were examined. Each variable except CO₂ is measured in µg/m³ units while CO₂ is measured in mg/m³. Table 2 provides a brief overview of the dataset. The Jaipur air quality dataset includes 24 variables, of which 19 are numerical, while the remaining are date and categorical variables. Additionally, the dataset contains over 5% missing values across different parameters. The histogram of Jaipur's AQI (Fig. 3) indicates that the AQI typically falls between 70 and 160. During the monsoon season, the AQI is generally lower, with a mean of 95, compared to summer and winter. The highest AQI recorded in the monsoon season is 233, while the lowest is 48. Winter sees the highest air pollution in Jaipur, with a mean PM₁₀ value of 143.15 and an AQI of 161.7. The Fig. 4 shows correlation between various pollutants. The AQI is strongly associated with PM₁₀ (r=.83) and PM_{2.5} (r=.83) while it is moderately associated with NO (r=.43) and NO₂ (r=.45). The Fig. 5 shows the feature importance graph for prediction of AQI.

2.2. Outlier detection methods

An observation that differs significantly from others and raises questions about its origin [40]. Outliers are observations whose values

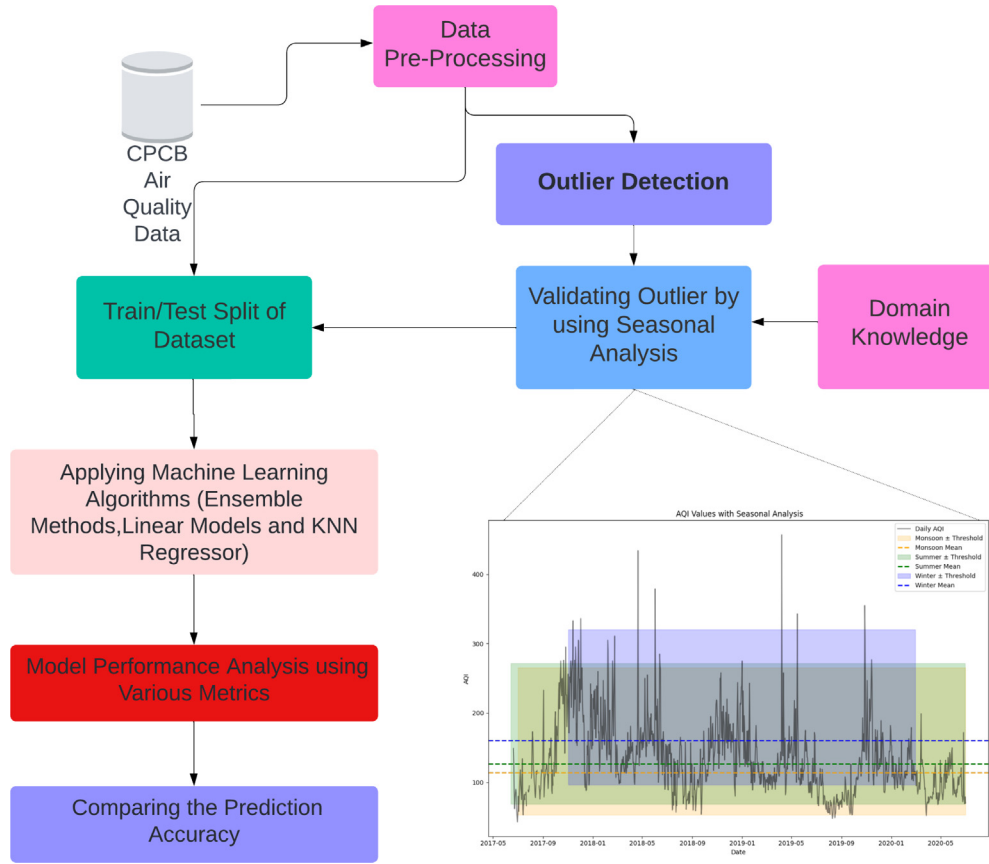


Fig. 2. Overall Flow for Prediction of Jaipur AQI.

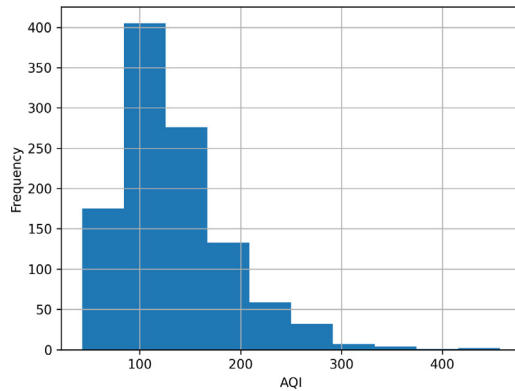


Fig. 3. AQI Histogram.

show a large deviation from mean value. The presence of an outlier can have significant influence on values of regression coefficients. Thus, it is important to identify the existence of outliers in the data. Outlier detection techniques can be classified into several categories based on different criteria i.e. Statistical Methods and Machine Learning-based Methods. In this study, the few statistical methods of outlier detection are employed to detect outliers.

2.2.1. InterQuartile Distance (IQD)

Interquartile distance (IQD), also called inter-quartile range (IQR) [41], is a measure of the distance between Quartile 1 (Q_1) and Quartile 3 (Q_3). IQD is a useful measure for identifying outliers in the data. Outlier is an observation which is far away (on either side) from the mean value of the data.

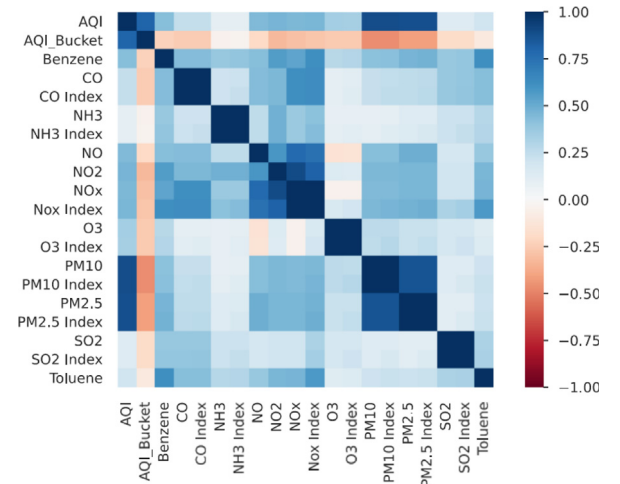


Fig. 4. Correlation of various pollutants with AQI.

The estimate of the lower quartile (the 0.25 quantile) is given by:

$$Q_1 = (1 - h)X[j] + hX[j + 1] \quad (1)$$

Where h is a weighting factor between 0 and 1, $X[j]$: This represents the value of the data point at position j in your ordered data set (from least to greatest). Since it is for the lower quartile (0.25 quantile), j typically corresponds to the position around the 25th percentile.

Letting $k = n - j + 1$, the estimate of the upper quartile is:

$$Q_3 = (1 - h)X[k] + hX[k - 1] \quad (2)$$

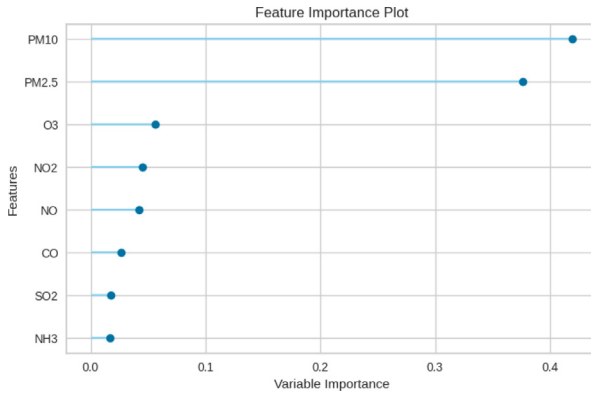


Fig. 5. Feature Importance Graph.

The estimate of the interquartile range is:

$$IQR = Q_3 - Q_1 \quad (3)$$

The IQR simply represents the difference between the estimated upper and lower quartiles. IQR can be used to define fences to capture the typical spread of the data. A common rule of thumb is to use 1.5 times the IQR as the "fence distance". Here's how the fences are calculated:

$$Lower Fence = Q_1 - (1.5 * IQR) \quad (4)$$

$$Upper Fence = Q_3 + (1.5 * IQR) \quad (5)$$

Fig. 6 illustrates the outliers in various pollutant concentrations and AQI identified using the Interquartile Range (IQR) method. The analysis revealed 21 outliers in the PM₁₀ data, with concentrations exceeding 253.8 µg/m³. For PM_{2.5}, PM₁₀ outliers were identified, while the AQI data exhibited 28 outliers. Additionally, a substantial number of outliers were detected in the concentrations of NO and NH₃, indicating significant deviations from typical values in these pollutants.

2.2.2. Robust Z-score

Z-score [42] is the standardized distance of an observation from its mean value. For the predicted value of the dependent variable Y, the Z-score is given by

$$z = (x - \mu) / \sigma \quad (6)$$

where x is the value of the data point, μ and σ are, respectively, the mean and the standard deviation of dependent variables estimated from the sample data. Any observation with a Z-score of more than 3 may be flagged as outlier and influential observations that may change the regression parameter values significantly. The standard z-score, derived from the mean and standard deviation, can be distorted by outliers. Such extreme values can increase the standard deviation, giving the impression that many data points are further from the mean than they really are. This distortion can cause the true extent of outliers' deviation from the "normal" range to be underestimated. Robust Z-Score approach [40] addresses this limitation by using alternative statistics that are less sensitive to outliers. It uses median and median absolute deviation.

$$z_{robust} = (x - M) / MAD \quad (7)$$

Where: z_{robust} : Represents the robust z-score, x : Represents the value of the data point you are interested, M : Represents the median of your data set and MAD : Represents the median absolute deviation. Typically, data points with robust z-scores beyond ± 2 or ± 3 are considered outliers. However, the specific threshold can vary depending on the application and the dataset. For PM_{2.5}, a total of 78 outliers were identified where the robust z-scores exceeded the thresholds of ± 3 .

Similarly, the PM₁₀ data revealed 55 outliers under the same robust z-score criteria. In the case of AQI, the analysis detected 89 outliers with robust z-scores beyond ± 3 . Fig. 7 is showing outliers in AQI detected by robust z score approach.

2.2.3. Mahalanobis distance

The Mahalanobis distance [43] serves as a measure of the disparity between a data point and a distribution. Mahalanobis distance is designed for multivariate data, meaning it considers multiple variables and their relationships (covariance) to determine how far a data point deviates from the central tendency. It is the distance between specific values of the independent variable (x) to the centroid of all observations of the explanatory variable. If Mahalanobis distance is significantly larger than the critical value from the chi-squared distribution with degrees of freedom equal to the number of variables, then x can be considered an outlier. The Mahalanobis distance is given by:

$$D_M(x) = \sqrt{(x - \mu)^T \Sigma^{-1} (x - \mu)} \quad (8)$$

Where x is vector of observation, μ is mean, Σ is covariance matrix, Σ^{-1} is inverse of covariance matrix and T denotes transpose of x . Following the calculation of Mahalanobis distances, the threshold for outlier detection is determined using the chi-squared distribution, considering the specified significance level and the number of features. Observations with Mahalanobis distances exceeding the square root of this threshold are classified as outliers. For the air quality dataset, which includes three features –PM_{2.5}, PM₁₀, and AQI – the Mahalanobis distance was computed. Using a lower significance level ($p = 0.01$) with three degrees of freedom, 44 observations were identified as outliers. In contrast, a higher significance level ($p = 0.05$) detected 68 observations as outliers.

2.2.4. Cook's distance

Cook's distance [44] measures how much the predicted value of the dependent variable changes for all the observations in the sample when a particular observation is excluded from the sample for the estimation of regression parameters. Cook's distance for simple linear regression is given by Aguinis et al. [45].

$$D_i = e_i^2 / \hat{\sigma}^2 p [h_{ii} / (1 - h_{ii})] \quad (9)$$

Where e_i^2 is residual for i th observation, $\hat{\sigma}^2$ is the mean squared error, p is number of parameters and h_{ii} is the leverage of i th observation. The Cook's distance method, which evaluates the influence of each observation on the regression model, tends to detect fewer outliers compared to the Mahalanobis distance approach. Table 3 is showing Mahalanobis distance and influence score. In this study, Cook's distance identified 32 observations as outliers, while the Mahalanobis distance method flagged 44 observations as outliers at a significance level of 0.01 and 68 observations at a significance level of 0.05. This discrepancy highlights the sensitivity of the Mahalanobis distance to the overall data distribution, making it a more stringent method for outlier detection in high-dimensional spaces. Specifically, for the air quality dataset with features PM_{2.5}, PM₁₀, and AQI, the Mahalanobis distance approach proved more effective in identifying anomalies that could significantly impact the model's performance.

2.3. Machine learning models

This research investigates the efficacy of linear models and ensemble models in predicting air quality in Jaipur, particularly in the presence of outliers. Among linear models, Linear Regression, Ridge Regression, and Lasso Regression were examined. Ensemble models, including Random Forest, Extra Tree Regressor, and Gradient Boost Regressor, were also evaluated. Additionally, the K-Nearest Neighbors (KNN) Regressor was implemented to enhance the prediction accuracy.

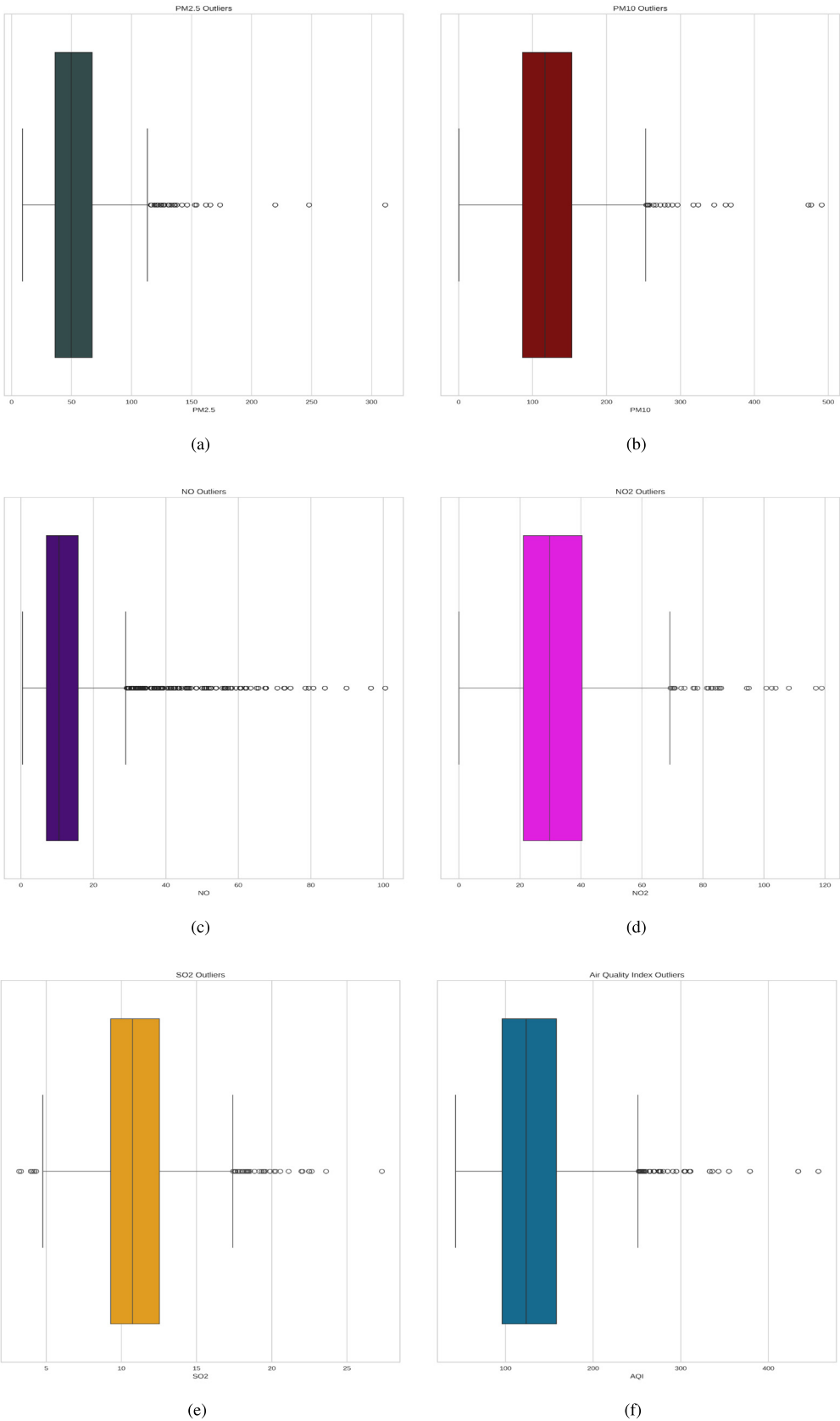


Fig. 6. Outliers in various pollutant concentrations identified by IQR: (a) PM_{2.5} (b) PM₁₀ (c) NO (d) NO₂ (e) SO₂ (f) AQI.

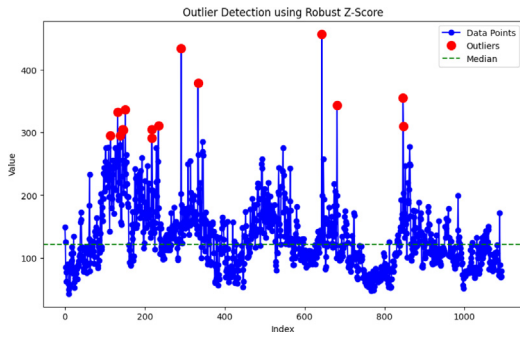


Fig. 7. Outlier detection by Robust Z score for AQI.

Table 3
Mahalanobis distance and influence score calculated.

PM2.5	PM10	AQI	Mahalanobis	Influence
54	123	133	1.47135796290122	0.000007
54	123	133	1.98476696129355	0.000007
54	123	133	1.53672515148771	0.000007
54	123	133	2.0557528130634	0.000007
54	123	133	1.29496397817197	0.000007
...	...	...	...	...
28.33	76.66	89	1.02496288572042	0.000058
23.76	67.44	75	1.18901775894494	0.000027
30.58	77.3	69	1.24019131510726	0.000019
30.03	64.26	79	1.10130338117977	0.000016
58.79	103.68	70	2.67735987771851	0.00266

2.3.1. Linear regression

Linear regression [46] is a statistical method used to model the relationship between a dependent variable and one or more independent variables by fitting a linear equation to observed data. It is a fundamental technique in statistical analysis and machine learning, widely utilized for predictive modeling and understanding the relationships between variables. The two key components of linear regression [47] are Dependent Variable (Y), Independent Variables (X). The dependent variable is variable that you aim to predict or explain. It is typically denoted as Y and is considered the response or outcome variable. Independent Variables used to predict the dependent variable. These are also known as predictor variables or features. In simple linear regression, there is only one independent variable (X), whereas in multiple linear regression, there can be multiple independent variables (X1, X2, X3, etc.). Linear regression assumes a linear relationship between the independent variable(s) and the dependent variable. This implies that a change in the independent variable(s) will result in a proportional change in the dependent variable.

The relationship between the independent and dependent variables is represented by a linear equation of the form:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon \quad (10)$$

In Eq. (10), Y is the dependent variable, $X_1, X_2, \dots, X_n$ are the independent variables. While β_0 is the intercept (the value of Y when all independent variables are zero) and $\beta_1, \beta_2, \dots, \beta_n$ are the coefficients (slopes) representing the change in Y for a one-unit change in each independent variable, holding other variables constant. ϵ is the error term, representing the difference between the predicted and actual values of Y. It accounts for unexplained variation in Y.

The aim of linear regression is to estimate the coefficients $\beta_0, \beta_1, \dots, \beta_n$ that minimize the difference between the observed values of the dependent variable and those predicted by the linear model. The predominant technique for estimating these coefficients is the least squares method [48], which minimizes the sum of the squared difference between the observed and predicted values of the dependent variable. Linear regression operates under the assumptions that

there is a linear relationship between the variables, the residuals (the differences between observed and predicted values) follow a normal distribution, and that there is homoscedasticity (constant variance) of residuals. Linear regression is extensively employed across various domains such as economics, finance, social sciences, engineering, and machine learning for tasks including prediction [49], forecasting, and elucidating relationships between variables.

2.3.2. Ridge regression

Ridge regression [50] is a regression analysis technique that is used to tackle the problem of multicollinearity (high correlation between predictor variables) and overfitting in linear regression models. It is similar to ordinary least squares (OLS) regression [48], but with the addition of a penalty term that shrinks the estimated coefficients towards zero. This penalty term helps to reduce the variance of the estimates, often at the expense of a small increase in bias. The equation of ridge regression

$$\hat{\beta}_{ridge} = (X'X + \lambda I_p)^{-1} X'Y \quad (11)$$

The objective function (equation (12)) for ridge regression is a modification of the least squares objective function. It includes an additional term to penalize large coefficients:

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p \beta_j^2 \quad (12)$$

Where y_i are the observed values of the dependent variable, $\hat{y}_i$ are the predicted values of the dependent variable and β_j are the coefficients of the predictor variables. λ is the regularization parameter that controls the strength of the penalty. When $\lambda=0$, ridge regression reduces to ordinary least squares regression. As λ increases, the magnitude of the coefficients decreases, which can help reduce variance but may introduce bias.

One of the primary motivations for employing ridge regression is to address multicollinearity, a condition in which predictor variables exhibit high correlations with one another. In the presence of multicollinearity, ordinary least squares (OLS) estimates can become unstable and highly sensitive to small variations in the data. Ridge regression mitigates this issue by reducing the variance of the coefficient estimates. This is achieved by introducing a small amount of bias into the estimates, which in turn reduces their variance. This trade-off can lead to improved predictive performance, particularly in high-dimensional datasets or those with correlated predictors. Ridge regression is extensively used in various fields, including statistics, machine learning, and data science, especially for regression problems where multicollinearity or overfitting is a concern. It is an effective tool for enhancing the stability and generalization performance of linear regression models.

2.3.3. Lasso regression

Lasso regression [51], which stands for Least Absolute Shrinkage and Selection Operator, is a method utilized for both variable selection and regularization in regression analysis. It is particularly advantageous when working with high-dimensional datasets that may contain numerous predictor variables, some of which could be irrelevant or redundant. Lasso regression addresses this by applying a penalty proportional to the absolute value of the regression coefficients, promoting the shrinkage of some coefficients towards zero. This not only reduces the complexity of the model but also inherently performs variable selection by setting certain coefficients to exactly zero, thus excluding the corresponding variables from the model. A key benefit of Lasso regression over Ridge regression is that Lasso not only reduces the magnitude of coefficients to address multicollinearity but also performs automatic variable selection by setting some coefficients to exactly zero, thereby simplifying the model and enhancing interpretability. The lasso objective function is given by:

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p |\beta_j| \quad (13)$$

One of the primary benefits of Lasso regression is its ability to automatically perform variable selection by setting some coefficients to zero. This is achieved through the penalty term in the Lasso regression objective function, which shrinks less important coefficients towards zero, effectively excluding them from the model. Lasso regression is widely used in various fields such as statistics, machine learning, and data science for tasks such as feature selection, predictive modeling, and regularization of models. It is particularly valuable when dealing with datasets with a large number of predictors or when interpretability of the model is important.

2.3.4. Random forest regressor

The concept of Random Forests was developed by Leo Breiman and Adele Cutler [52]. This method combines multiple decision trees to improve the predictive performance and control overfitting. It effectively minimizes overfitting and enhances the model's generalization ability. In the Random Forest regressor [53], the prediction for a given instance x_i is computed as the average of predictions from individual decision trees. Let $f_m(x_i)$ denote the prediction of the m th decision tree for input x_i , where m ranges from 1 to the total number of trees in the forest. The final prediction $\hat{y}_i$ for x_i is then calculated as the average of these individual predictions:

$$\hat{y}_i = 1/M \sum_{m=1}^M f_m(x_i) \quad (14)$$

Where M is the total number of trees in the forest. Each decision tree is built on a bootstrapped sample of the training data, ensuring diversity among the trees. Moreover, at each split in a decision tree, a random subset of features is considered, reducing the correlation among trees and improving the robustness of the model. This ensemble approach helps in capturing complex nonlinear relationships in the data while mitigating overfitting.

Random forest's simplicity of implementation, scalability, and robustness to outliers contribute to its widespread adoption in both academic research and industrial applications. Its versatility extends to handling large datasets with numerous features, making it well-suited for applications in finance, ecology, and climate modeling, among others.

2.3.5. Gradient boosting regression

Gradient Boosting Regression [54] is a powerful machine learning technique used for regression tasks. It belongs to the class of ensemble learning methods, which combine the predictions of multiple individual models (often called weak learners or base learners) to produce a stronger predictive model. Gradient Boosting Regression builds models sequentially, with each new model attempting to correct the errors made by the previous ones. The base learner used in Gradient Boosting Regression is typically decision trees. However, other types of weak learners can also be used. Decision trees are often used due to their ability to capture non-linear relationships and interactions between features. Gradient Boosting Regression builds a sequence of decision trees. Each tree is trained on the errors (residuals) made by the previous trees in the sequence. The algorithm minimizes the loss by following the gradient (direction of steepest descent) of the loss function with respect to the predicted values. Each new tree is trained to predict the negative gradient of the loss function, effectively reducing the residual error. Gradient Boosting Regression introduces a learning rate parameter (also known as shrinkage or step size), which controls the contribution of each tree to the final prediction. A smaller learning rate results in slower learning but can improve the generalization performance of the model by reducing overfitting. The training process continues until a predefined stopping criterion is met. This could be a maximum number of trees to build, reaching a minimum level of error, or other criteria specified by the user. Gradient Boosting Regression combines the predictions of all the individual trees in the sequence. The final prediction is the sum of the predictions of all the trees, weighted by the learning rate.

Gradient Boosting Regression is robust to outliers in the data due to the sequential nature of model building. Gradient Boosting Regression can capture complex non-linear relationships between features and the target variable, making it suitable for a wide range of regression problems [55]. Gradient Boosting Regression is widely used in various applications such as finance, healthcare, and marketing, where accurate predictions of continuous outcomes are required. Popular implementations of Gradient Boosting Regression include XGBoost, LightGBM, and CatBoost. These libraries offer highly optimized and efficient implementations of Gradient Boosting algorithms.

2.3.6. Extra trees regressor

The Extra Trees regressor [56], an extension of the Random Forest algorithm, stands as a potent tool in the realm of regression tasks. It shares the foundation of ensemble learning, harnessing the collective wisdom of multiple individual models to forge a more robust predictive framework. The Extra Trees regressor primarily employs decision trees as its base learner. These decision trees excel in capturing intricate non-linear relationships and feature interactions present in the data. While Random Forest typically considers a subset of features at each split, the Extra Trees regressor goes a step further by randomly selecting feature thresholds, without the need for a deterministic optimization process. This randomization enhances diversity among the trees and promotes exploration of different regions of the feature space. The Extra Trees regressor aggregates the predictions from all the individual trees in the ensemble to predict outcomes for new data points. The final prediction is often computed as the average or weighted average of these individual predictions, depending on the variant of the algorithm used.

The inherent diversity introduced by randomized feature selection and splitting renders the Extra Trees regressor resilient to noise and outliers in the data, enhancing its robustness. The randomized nature of feature selection and splitting enables faster training times compared to traditional decision tree algorithms, making it well-suited for large datasets. Extra Trees regression finds widespread application across various domains where accurate predictions of continuous outcomes are imperative.

2.4. Parameter evaluation and model selection

In this study, R^2 Score, Mean Squared Error, and Root Mean Squared Error (RMSE) metrics have been used. These validation metrics are used to determine whether regression models are accurate or misleading.

2.4.1. Mean Absolute Error (MAE)

MAE [57] for model determines how well the model's predictions match the actual values on average. MAE measures the average absolute differences between the predicted values and the actual values. Lower MAE values indicate better performance, with a MAE of 0 representing a perfect fit. The MAE can be evaluated by:

$$MAE = \frac{\sum_{i=1}^n |Y_i - \hat{Y}_i|}{n} \quad (15)$$

Where Y_i is actual value and $\hat{Y}_i$ is predicted value.

2.4.2. Root mean square error (RMSE)

When evaluating a model using RMSE, the aim is to determine how well the model's predictions match the actual values on average, with greater emphasis on large errors compared to MAE. RMSE is the square root of the average squared differences between the predicted values and the actual values. This metric retains the properties of MSE [49], where larger errors have a disproportionately greater impact, but it provides an error metric in the same units as the original data, making it more interpretable. Lower RMSE values indicate better performance, with an RMSE of 0 representing a perfect fit.

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (\hat{Y}_i - Y_i)^2}{n}} \quad (16)$$

Table 4
Seasonal values of various pollutants.

SEASON		$PM_{2.5}$	PM_{10}	NO	NO_2	AQI
Mean	MONSOON(July to September)	37.79	89.12	9.36	23.15	95
Max		92.75	214.16	42.62	58.87	233
Min		11.5	0.16	0.44	3.67	48
Mean	WINTER(November to February)	70.2	143.15	21.84	42.37	160.5
Max		311.15	317.48	100.51	118.98	336
Min		28.2	49.68	2.39	4.9	80
Mean	SUMMER(March to June)	47.53	123.89	9.86	27.02	126.83
Max		126.39	491.01	50.58	60.46	457
Min		9.12	14.47	2.55	0.1	43

Table 5
Number of outliers detected by various methods.

	IQR	Robust Z score	Mahalanobis distance	Cook's distance
PM2.5	10	78	NA ^a	NA ^a
PM10	21	55	NA ^a	NA ^a
AQI	28	89	68	32

^a Work only for multivariate outlier detection.

2.4.3. R^2 score

When evaluating a model using the coefficient of determination (R^2), also known as R-squared, the aim is to determine how well the model's predictions fit the actual values relative to a simple baseline model. R^2 measures the proportion of the variance in the dependent variable that is explained by the independent variables in the model. Higher R^2 values indicate better performance [58], with a maximum value of 1 indicating a perfect fit. R^2 helps in optimizing your model for accurate predictions on unseen data while accounting for the variance explained by the model.

$$R^2 = 1 - \frac{\text{sum squared regression (SSR)}}{\text{total sum of squares (SST)}}, \quad (17)$$

$$R^2 = 1 - \frac{\sum(Y_i - \hat{Y}_i)^2}{\sum(Y_i - \bar{Y})^2} \quad (18)$$

3. Result analysis

This section delves into the seasonal analysis of outliers, identified using the outlier detection methods discussed in Section 2.2. The performance of various machine learning models is evaluated based on Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score, as detailed in Section 2.4. The analysis encompasses both linear models – namely, linear regression, lasso regression, and ridge regression – and ensemble models, including random forest, gradient boosting, and extra trees regressor. Additionally, the k-nearest neighbors (KNN) regressor is also employed. To thoroughly assess the impact of outliers on model performance, each model is applied to two distinct datasets: one containing the original data with outliers and another with outliers removed. This dual approach enables a comprehensive evaluation of how outlier presence affects prediction accuracy.

3.1. Seasonal analysis

Seasonal analysis for air quality prediction is crucial due to the significant variations in pollutant levels driven by seasonal changes [58], [59]. Factors such as temperature, humidity, wind patterns, and human activities fluctuate throughout the year, influencing the concentration and dispersion of pollutants. For instance, higher temperatures in summer can increase the formation of ground-level ozone [60], while winter heating can elevate particulate matter levels. The impact of seasonal variation in air quality is observed in many studies [61–63]. Understanding these seasonal patterns enables more accurate predictions of air quality, facilitating better public health advisories and environmental management. Additionally, identifying and accounting

for seasonal outliers can improve the robustness of predictive models, ensuring they perform reliably across different times of the year. Although numerous frameworks [64–66] have been proposed for air quality prediction, a combination of multiple outlier detection methods integrated with seasonal validation has not been used in framework, as indicated by the literature review. This study integrates multiple outlier detection methods with season validation in air quality prediction framework.

The seasonal maximum, minimum and mean value is shown in Table 4. Table 5 is showing the number of outliers detected by various statistical methods. In our analysis, the robust z-score method identified the highest number of outliers, while the Interquartile Range (IQR) method identified the fewest. Although eliminating a large number of outliers can enhance the R^2 score and accuracy for the training data, it may compromise the model's performance on unseen data. In this paper a seasonal validation is done for outliers flagged by all methods. During the winter season, higher values are more acceptable, whereas in the summer and monsoon seasons, peak values for $PM_{2.5}$, PM_{10} , and AQI tend to be lower compared to winter. This seasonal variation is crucial for accurate environmental modeling and prediction. For each season, daily readings from every monitoring station were examined. The highest $PM_{2.5}$ and PM_{10} values recorded during each season were selected as the representative peak concentrations. Exceptional high values resulting from biomass burning and wildfires in winter were marked as outliers. Additionally, during the study period, several days were identified as being affected by sandstorms. These sandstorm-affected days were not included when determining peak values for marking outliers.

Seasonal extreme values should be defined based on the expected range for a given season. High levels of air pollution are common during the winter for some regions, hence higher PM and AQI results should not be considered outliers. During the summer months, pollution levels are generally low. Recognizing the particular environmental factors involved is essential for accurately detecting and eliminating outliers in pollution data. The formal model for outlier removal is outlined below:

1. Let $Summer_{max}(AQI)$ and $Summer_{min}(AQI)$ represent the maximum and minimum values of the AQI in summer season. Similarly $Winter_{max}(AQI)$, $Winter_{min}(AQI)$, $Monsoon_{max}(AQI)$, and $Monsoon_{min}(AQI)$ represents maximum and minimum AQI values for winter and monsoon.
2. The seasonal threshold (for example summer threshold: $Summer_{threshold}(AQI)$) is defined based on extreme values observed for a particular season. This can be based on the standard

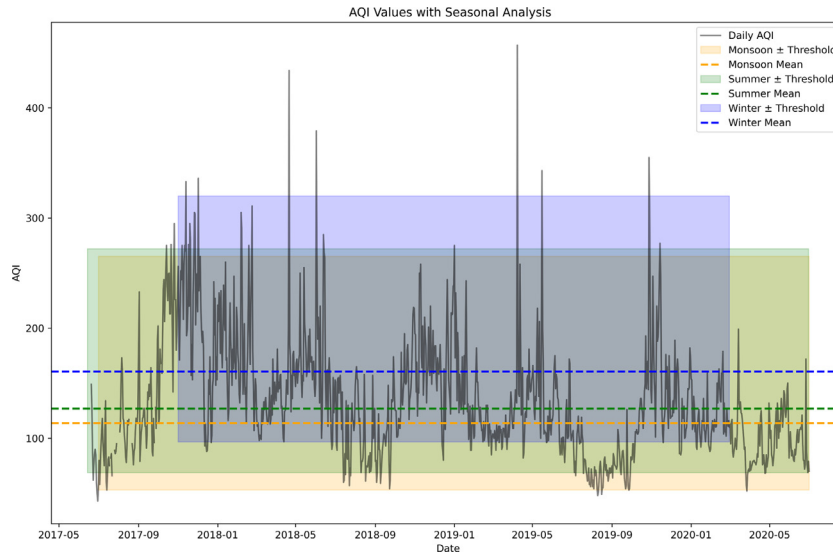


Fig. 8. Seasonal analysis of outliers.

Table 6
Seasonal analysis of outlier by using statistics and domain knowledge.

Season	Count	Mean	Std Deviation	Min	25.00%	50.00%	75.00%	Max	Seasonal threshold
Monsoon	264.00	95.38	31.28	48.00	71.75	88.00	114.25	233.00	192.34
Summer	377.00	126.83	48.39	43.00	97.00	118.00	149.00	457.00	276.83
Winter	361.00	160.55	53.16	80.00	116.00	153.00	193.00	336.00	325.35

deviation of the seasonal values.

$$Summer_{threshold}(t) = \mu_{Summer} + \kappa \cdot \sigma_{Summer} \quad (19)$$

where:

- μ_{Summer} is the mean of the AQI values in summer.
- σ_{Summer} is the standard deviation of the AQI in summer.
- κ is a tuning parameter that controls how extreme AQI values need to be to classify as extreme but not outliers. Choosing κ requires domain knowledge.

Fig. 8 shows seasonal analysis of AQI values by using statistics. The final outlier numbers after applying above are 14 in AQI (Monsoon $AQI > 192$, Summer $AQI > 276$, and Winter $AQI > 325$) after considering seasonal statistics (Table 6). For an AQI reading of 234, traditional methods might classify it as a typical value. However, since this reading occurs during the monsoon season, applying seasonal analysis would identify it as an outlier. By adjusting thresholds according to seasonal variations, this approach effectively captures the true characteristics of the data while remaining resilient to anomalies.

3.2. Comparative analysis of machine learning methods

This study employs linear models, ensemble methods, and K-Nearest Neighbors (KNN) regression to predict the Air Quality Index (AQI) in the presence of outliers. Linear models such as Linear Regression and Ridge Regression are significantly impacted by outliers. For instance, Linear Regression exhibits a Mean Absolute Error (MAE) of 15.4086, a Root Mean Square Error (RMSE) of 25.2014, and an R^2 score of 0.7451 in the presence of outliers (Table 7). After removing outliers, the performance improves to an MAE of 13.2774, an RMSE of 17.3249, and an R^2 score of 0.8635. Similarly, Ridge Regression shows an MAE of 15.192, an RMSE of 25.0386, and an R^2 score of 0.7495 for the baseline model, which improves to an MAE of 3.4542, an RMSE of 17.1244, and an R^2 score of 0.8708 after outlier removal. Notably, Lasso Regression benefits the most from outlier removal, with its R^2 score improving

from 0.58 to 0.83, representing an accuracy improvement of over 25%. Lasso regression uses the L1 norm for regularization, which penalizes the sum of the absolute values of the coefficients. This might be reason for poor performance in presence of outliers. Fig. 9 compares various models' performance by using R^2 score. The KNN regressor also exhibited benefits from outlier removal. It has given MAE = 17.5577, RMSE = 25.8062, R^2 score = 0.7432 for baseline model and MAE = 15.7903, RMSE = 21.1012 and R^2 score = 0.8137. KNN's sensitivity to outliers stems from its dependence on Euclidean distances, which can disproportionately weight anomalous data points. Removing outliers refines the neighborhood for predictions, improving accuracy.

All ensemble models show robustness against outliers. Gradient Boosting Regressor for learning_rate = 0.1, alpha=0.9 and 100 boosting stages MAE = 13.2523, RMSE = 19.6805 and R^2 score = 0.8382 for baseline model, while after outlier removal it exhibited MAE = 13.2925, RMSE = 18.4365 and R^2 score = 0.8453. The Random Forest Regressor exhibited a MAE of 13.4888, a Root Mean Square Error of 19.4115, and an R^2 score of 0.8481 for the baseline model. After refinement, the model showed slight improvement with an MAE of 13.0231, an RMSE of 18.4451, and an R^2 score of 0.8464. The Extra Trees Regressor performed best with a MAE of 12.6765, RMSE of 17.8452, and an R^2 score of 0.8737 for the baseline model. The randomness in feature selection and splitting makes Extra Trees less sensitive to outliers, yielding superior performance, especially after refinement. After refinement, the model showed improved performance with an MAE of 11.9161, an RMSE of 16.1660, and an R^2 score of 0.8884. The randomness in feature selection and split decisions in Extra Trees diminishes the influence of outliers, leading to a more robust model in case of noisy dataset. The prediction error of baseline model and after removal of outliers is shown in Fig. 10 (see Table 7).

4. Conclusion

This study collected Air Quality data for Jaipur city, India, from the Central Pollution Control Board (CPCB) and applied several data

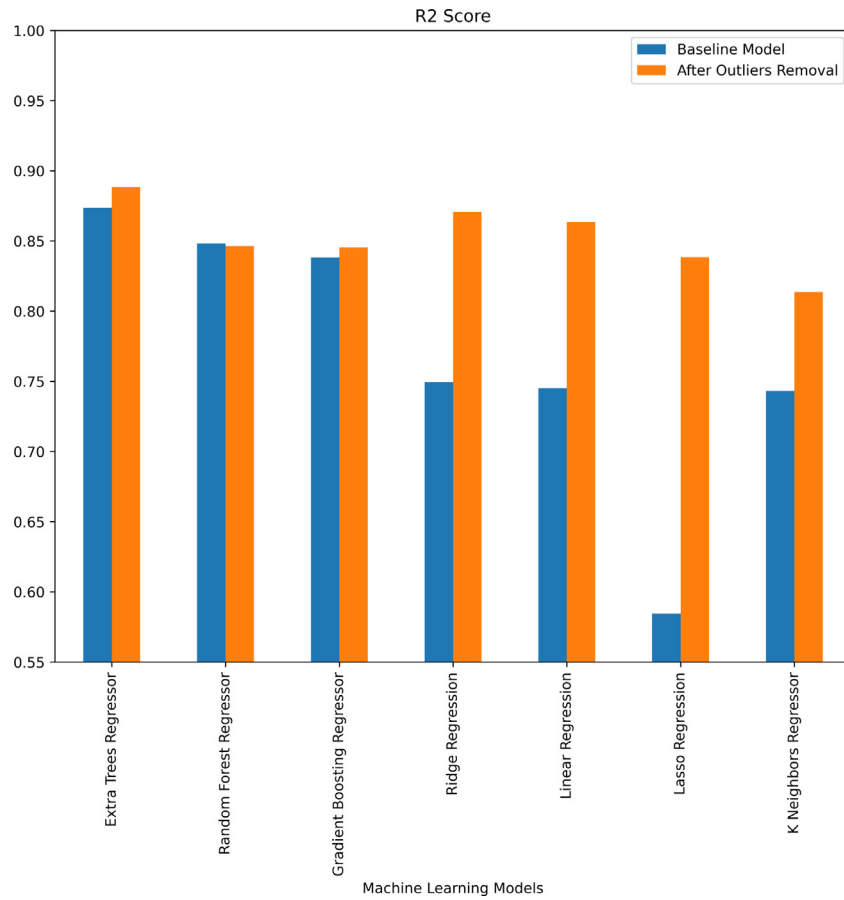


Fig. 9. Performance of various machine learning models for AQI prediction.

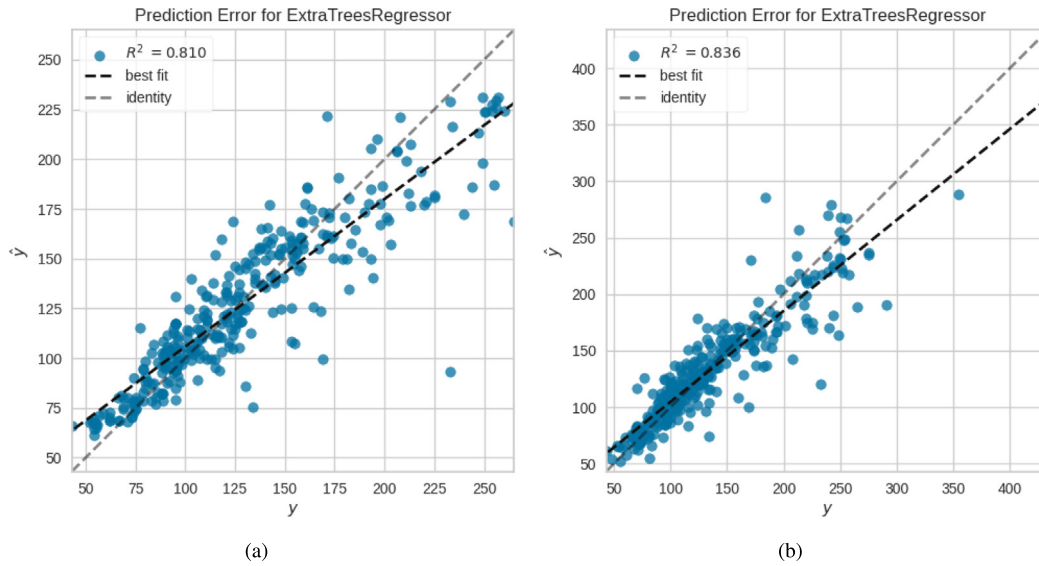


Fig. 10. Prediction Error for Extra Tree Regressor (a) Before removal of outliers (b) After removal of outliers.

pre-processing methods to normalize the data and manage missing values. Outlier analysis using statistical techniques, combined with seasonal validation, provided a comprehensive understanding of the data's anomalies. The proposed architecture, depicted in Fig. 2, which mainly focus on seasonal validation of outliers can employ various linear models (including Linear Regression, Lasso, and Ridge Regression) and ensemble methods (such as Random Forest, Extra Trees, and

Gradient Boosting) to predict the AQI of Jaipur city. A comparative analysis of these models revealed that ensemble methods are more robust to outliers compared to linear models. Linear models like Linear Regression and Ridge Regression showed significant performance improvements after outlier removal. For instance, Linear Regression's R^2 score improved from 0.7451 to 0.8635, and Ridge Regression's R^2 score increased from 0.7495 to 0.8708. Lasso Regression benefited the most,

Table 7
Performance of various machine learning algorithms.

Model	Before Outliers Removal			After Outliers Removal		
	MAE	RMSE	R ²	MAE	RMSE	R ²
Extra Trees Regressor	12.6765	17.8452	0.8737	11.9161	16.166	0.8884
Random Forest Regressor	13.4888	19.4115	0.8481	13.0231	18.4451	0.8464
Gradient Boosting Regressor	13.2523	19.6805	0.8382	13.2925	18.4365	0.8453
Ridge Regression	15.192	25.0386	0.7495	13.4542	17.1244	0.8708
Linear Regression	15.4086	25.2014	0.7451	13.2774	17.3249	0.8635
Lasso Regression	20.7254	31.1252	0.5844	15.2193	19.6624	0.8384
K Neighbors Regressor	17.5577	25.8062	0.7432	15.7903	21.1012	0.8137

with its R^2 score rising from 0.58 to 0.83, representing an accuracy improvement of over 25%.

Ensemble methods consistently demonstrated robustness against outliers. Ensemble models like Random Forest or Gradient Boosting do not rely on linear assumptions. They can capture complex, non-linear relationships between variables, which allows them to be more flexible and robust in handling outliers. The Gradient Boosting Regressor, with a learning rate of 0.1, alpha of 0.9, and 100 boosting stages, showed slight improvements in performance metrics after outlier removal. The Random Forest Regressor exhibited minor improvements as well. However, the Extra Trees Regressor outperformed all other models, achieving the best results with an MAE of 12.6765, an RMSE of 17.8452, and an R^2 score of 0.8737 for the baseline model. After removal of outliers, the Extra Trees Regressor improved to an MAE of 11.9161, an RMSE of 16.1660, and an R^2 score of 0.8884.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

- All authors have participated in (a) conception and design, or analysis and interpretation of the data; (b) drafting the article or revising it critically for important intellectual content; and (c) approval of the final version.
- This manuscript has not been submitted to, nor is under review at, another journal or other publishing venue.
- The authors have no affiliation with any organization with a direct or indirect financial interest in the subject matter discussed in the manuscript

Data availability

Data will be made available on request.

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